

Michael Grier

Software developer

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GitHub: github.com/michael-grier

Full-stack software developer with five years at TransAlta building renewable-energy reporting tools and business process automation. Co-led a reporting and data-governance platform and maintained a wind, solar, and hydro operations app, working across React interfaces, APIs, databases, and Azure infrastructure. Independent projects include ecommerce, a native mobile app, and browser-based developer tools.

Experience

Software Developer

TransAlta, Calgary | June 2023 to June 2026

AthenaOS | Production reporting and data governance

- Co-led development of a Next.js and TypeScript platform that lets SQL-fluent analysts assemble wind and hydro reporting dashboards without writing React.
- Built reusable ECharts cards and chart presets, plus a data-catalog workflow with draft submissions, multi-stage review, rework, and audit trails.
- Implemented Microsoft Entra ID SSO, PostgreSQL-backed sessions, and distributed locking for token refresh across application instances.
- Built Terraform and GitHub Actions delivery workflows across development, UAT, and production, including release promotion, hotfixes, and rollback.

GenOS | Wind, solar, and hydro operations

- Migrated event data from REST to GraphQL and replaced a 1,700-line shared React component with focused wind, solar, and hydro modules.
- Built reusable mobile and tablet modals for event details and monitoring graphs; established Docker builds and Azure Pipelines across three environments.
- Fixed SQL Server connection leaks and strengthened JWT verification. Corrected capacity values, event filtering, and multi-unit CSV exports used in NERC reporting.

Junior Automation Developer

TransAlta, Calgary | June 2021 to May 2023

Developed and maintained Blue Prism automations, from stakeholder requirements and reusable process components to testing, scheduled execution, and troubleshooting deployed bots.

Freelance Software Developer

Self-employed, Calgary | February 2021 to June 2021

Led frontend design and development for a Django web project and supported Unity mobile game development.

Frontend Web Developer Intern

Invigulus, Remote | November 2020 to December 2020

Built a React application connected to a REST API and NoSQL database, giving employees a web interface for data entry and retrieval.

Selected projects

Independent development

Skateboard Ecommerce Platform

Checkout, inventory, and order management for an independent skateboard brand

Built checkout with server-validated pricing and atomic inventory reservations, retry-safe payment processing, and order management. Added end-to-end tests for price tampering, duplicate payments, and stock release.

Next.js, Stripe, Neon PostgreSQL, Drizzle, Cloudflare R2, Playwright

YYC Skate Spots

Community skateboarding map for Calgary

Built a mobile spot book with map clustering, location filters, photo submissions, favourites, and moderation. Added share links with a web fallback and prepared the iOS release configuration and store submission.

React Native, Expo, Convex, Clerk, Google Maps, EAS

Code Trainer

Interactive TypeScript interview practice

Built a browser-based coding workspace with isolated execution, TypeScript checking, and automated grading. Combined lessons and exercises with local guest progress and account-based synchronization across devices.

React, Vite, Monaco Editor, Web Workers, Convex, Better Auth, Vitest

Technical skills

Frontend and mobile: TypeScript, JavaScript, React, Next.js, React Native, Expo, Tailwind CSS, Material UI, shadcn/ui, ECharts

Backend and data: Node.js, Express, REST, GraphQL, PostgreSQL, SQL Server, Databricks, Convex, Drizzle, Zod, Python, FastAPI

Cloud and integrations: Azure, Terraform, Docker, GitHub Actions, Azure Pipelines, Stripe, Cloudflare R2, Microsoft Entra ID, Clerk, Better Auth

Testing and tools: Playwright, Vitest, Jest, React Testing Library, Bun, Blue Prism

Education

Object-Oriented Software Development Certificate

Southern Alberta Institute of Technology | 2020 · GPA 3.98

Bachelor of Education

University of Calgary | 2016

Bachelor of Arts

University of Calgary | 2011